

App Development Project Report

App Dev Project Report

1. Student Details

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About Me: I am a student of time. Having a keen knowledge in computing, computer hardware, networking, cloud computing and open source software. arch btw...

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a web-based application that allows users/students to **apply, manage, and find suitable job profiles**. And also companies can **enlist, shortlist and have an organized list** of drives and applications.

Approach:

The app was built using Flask as the backend framework with a modular structure, and Vue for the frontend framework with SPA development approach for faster and cleaner look.

3. AI/LLM Declaration

I used **Claude (Sonnet 5)** to assist in writing the styling and ui cleanliness with bootstrap components and UI uniqueness throughout the app and also in bug fixes and tools initialization.

The extent of AI/LLM usage is around **10–15%**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Celery	For chat webhook synchronization
Bootstrap 5	Frontend styling and responsive design
Vue	Main Frontend framework
Vue-Router	For router usage in SPA
Axios	Asynchronous api calls to internal server
Flask-JWT	User authentication and session management
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram

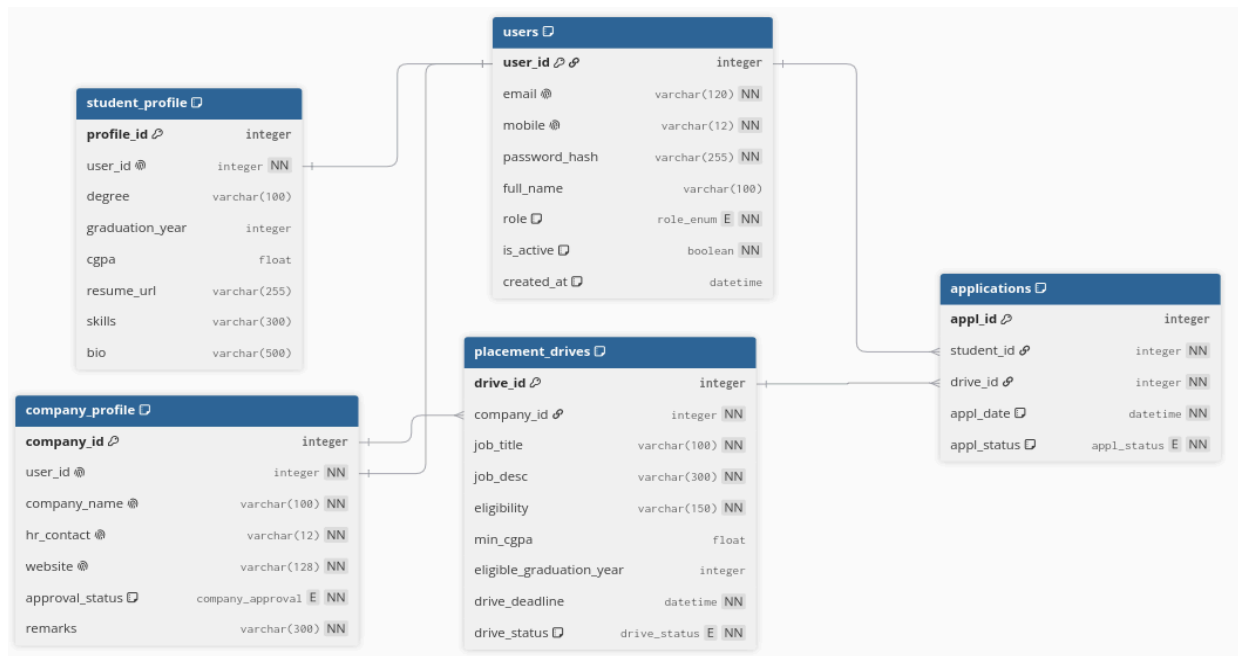
Tables:

1. **User** — all user profile details (user_id, email, mobile, full_name, role, is_active, created_at, password)
2. **StudentProfile** — specific student user profile details (profile_id, user_id, degree, graduation_year, cgpa, resume_url, skills, bio)
3. **CompanyProfile** — stores company profile (company_id, user_id, company_name, hr_contact, website, approval_status, remarks)
4. **PlacementDrives** — stores drive details (drive_id, company_id, job_title, job_desc, eligibility, min_cgpa, eligible_graduation_year, drive_deadline, drive_status)
5. **Applications** — stores applications details (appl_id, student_id, drive_id, appl_date, appl_status)

Relationships:

- One-to-One → User → StudentProfile
- One-to-One → User → CompanyProfile
- One-to-Many → CompanyProfile → PlacementDrives
- One-to-One → Applications → Users

- One-to-One → Applications → PlacementDrives



6. API Resource Endpoints

Endpoint	Method	Description
/api/auth/register	POST	Create and add user to db and generate access token for session
/api/auth/register-company	POST	Solely used for creating company profile and also user details to db
/api/auth/login	POST	Check db for authentication of user
/api/auth/logout	POST	Unsets jwt token, and by so deauthenticate user
/api/profile	GET	Fetching profile details of user
/api/profile	PUT	Updating profile details of user
/api/dashboard	GET	Showing simple stats about user

<code>/api/drives</code>	GET	Fetches list of approved drives
<code>/api/applications</code>	GET	Fetches list of applied drives of user
<code>/api/admin/stats</code>	GET	Fetches overall db statistics

More APIs Available in the app, will include in a `api.yaml`

7. Architecture and Features (optional)

Architecture Overview:

- **backend/app.py** – main Flask application entry point
- **backend/database** – database models using SQLAlchemy
- **backend/routes** – Flask Blueprints for auth, user, admin and company routes
- **backend/utils** – Extra utility scripts for the app
- **frontend/public** – contains app favicon and svg files
- **frontend/src/App.vue** – Main frontend entry
- **frontend/src/main.js** – Main JS file for the frontend
- **frontend/src/router** – Contains frontend routing
- **frontend/src/store** – Contains frontend authentication
- **frontend/src/views** – Contains frontend views of different routes
- **frontend/src/components** – Contains frontend modular components to be reused

Implemented Features:

Admin, Company, User registration and login
 JWT authenticated sessions
 Beautiful UI UX throughout app
 Search options for different views
 Responsive UI built with Bootstrap
 Secure password hashing and user session handling

Additional Features:

- Google Chat Notification System
 - AI-generated weekly summary insights (optional extension, yet to be implemented)
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8. Video Presentation

Drive Link:

 2026-07-14 22-10-55.mp4

<https://drive.google.com/file/d/1ji8yDFTisgorRvfa2tOaD0UBW2VHgXZn/view?usp=sharing>
